

The Maximum Number of Approximately Orthogonal Unit Vectors in $\mathbb{R}^d$

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Abstract—This paper investigates the problem of generating approximately independent random variables from given randomness. We begin with the simplest case of linear transformations of independent Gaussian random variables and extend our analysis to arbitrary distributions. The main focus is on the maximum number of approximately orthogonal unit vectors in $\mathbb{R}^d$ and the implications for generating independent random variables.

Index Terms—almost orthogonal vectors, equiangular lines, random variables

I. INTRODUCTION

In this project, we will explore how many we can generate approximately independent random variables from given randomness, and how we can generate them. we will begin at the simplest case, which is *linear* transformation of independent Gaussian random variables. Then the problem is reduced to the maximum number of approximately orthogonal unit vectors in $\mathbb{R}^d$, and we will extend it to equiangular lines and approximately equiangular lines..

A. Linear Transformation of Independent Gaussian Random Variables

For a fixed dimension n , let $\mathbf{x} = (x_1, x_2, \dots, x_n)$ be a vector of independent Gaussian random variables with mean 0 and variance 1. Consider a linear transformation $\mathbf{A} \in \mathbb{R}^{m \times n}$ applied to $\mathbf{x}$ to generate m random variables $\mathbf{y} = (y_1, y_2, \dots, y_m)$, where:

$$\mathbf{y} = \mathbf{A}\mathbf{x}.$$

Each row $\mathbf{a}_i$ of $\mathbf{A}$ is a unit vector, i.e., $\|\mathbf{a}_i\| = 1$ for $i = 1, \dots, m$.

Since $\mathbf{x}$ consists of i.i.d. standard Gaussian variables, the covariance matrix of $\mathbf{y}$ is given by:

$$\text{Cov}(\mathbf{y}) = \mathbf{A}\mathbf{A}^T = \begin{bmatrix} \mathbf{a}_1^T \mathbf{a}_1 & \mathbf{a}_1^T \mathbf{a}_2 & \cdots & \mathbf{a}_1^T \mathbf{a}_m \\ \mathbf{a}_2^T \mathbf{a}_1 & \mathbf{a}_2^T \mathbf{a}_2 & \cdots & \mathbf{a}_2^T \mathbf{a}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_m^T \mathbf{a}_1 & \mathbf{a}_m^T \mathbf{a}_2 & \cdots & \mathbf{a}_m^T \mathbf{a}_m \end{bmatrix}.$$

The diagonal entries satisfy $\mathbf{a}_i^T \mathbf{a}_i = 1$, while the off-diagonal entries $\mathbf{a}_i^T \mathbf{a}_j$ ($i \neq j$) measure the correlation between y_i and y_j .

For a given $\epsilon \in (0, 1)$, we say $\mathbf{y}$ is **approximately independent** in the ϵ -sense if:

$$|\mathbf{a}_i^T \mathbf{a}_j| < \epsilon \quad \forall i \neq j.$$

This condition ensures that the correlations between distinct y_i and y_j are small. The problem reduces to finding the maximum number M of unit vectors in $\mathbb{R}^n$ such that the absolute inner product between any two distinct vectors is less than ϵ .

II. THE MAXIMUM NUMBER OF APPROXIMATELY ORTHOGONAL UNIT VECTORS IN $\mathbb{R}^d$

A. Proving Existence from Probabilistic Arguments

Let $v_1, v_2, \dots, v_m$ be unit vectors in $\mathbb{R}^d$ such that the absolute value of the inner product between any two distinct vectors is at most ϵ , i.e., $|\langle v_i, v_j \rangle| \leq \epsilon$ for all $i \neq j$. We want to determine the maximum possible value of m (denoted by M) in terms of a function of d and ϵ .

In our previous discussion on Johnson-Lindenstrauss lemma, we have seen that to put n vectors in $\mathbb{R}^m$, we only need $O(\epsilon^{-2} \log n)$ dimensions:

Theorem 1 (Johnson-Lindenstrauss Lemma). *There exists C such that the following holds. Let $\epsilon \in (0, 1)$. For any set of n points $v_1, \dots, v_n$ in $\mathbb{R}^m$. Then for any $d \geq C\epsilon^{-2} \log n$. Let P be a random matrix in $\mathbb{R}^{d \times m}$, and each coordinate sampled from Gaussian with zero mean and $\frac{1}{d}$. Then with probability at least $1 - \frac{1}{n}$,*

$$(1 - \epsilon)\|v_i - v_j\|^2 \leq \|Pv_i - Pv_j\|^2 \leq (1 + \epsilon)\|v_i - v_j\|^2 \quad (1)$$

Proof. The proof can be seen in the textbook. $\square$

By intuition, the information of m dimensional space can be compressed to $d (<< m)$ dimensional space. Consider a special case of projecting the set of standard orthonormal basis $(e_i)_{i=1}^m$ of $\mathbb{R}^m$. Then Pe_i is the i -th column of P . Now we argue that there exists such a P that the m columns of

P are approximately orthogonal in $\mathbb{R}^d$. Consider m is large enough for $1 - \frac{1}{m}$ to be strictly positive, by 1 there must exist such a projection matrix P such that the distances between Pv_i and Pv_j $i \neq j$ are preserved up to a factor of $1 \pm \epsilon$. If we want the norm of each Pv_i also be preserved, we only need to take extra union bound for m v_i s, which does not change the order that $d \geq O(\frac{\log m}{\epsilon^2})$. Therefore the inner product of Pv_i and Pv_j $i \neq j$ is preserved to be nearly zero ($2\langle Pv_i, Pv_j \rangle = \|Pv_i\|^2 + \|Pv_j\|^2 - \|Pv_i - Pv_j\|^2$). To sum up, to put m orthonormal vectors in $\mathbb{R}^m$ we only need $O(\frac{\log m}{\epsilon^2})$ dimensions. Based on this intuition, we conjecture that the maximum number of approximately independent (orthogonal) unit vectors in $\mathbb{R}^d$ is $O(e^{\epsilon^2 d})$.

B. A More Accurate Lower Bound from Probabilistic Arguments

There is a small gap from our main focus: the column vectors of P are not unit vectors. Fortunately, we have a similar probabilistic argument for unit vectors using Hoeffding's inequality, which can be found in [1].

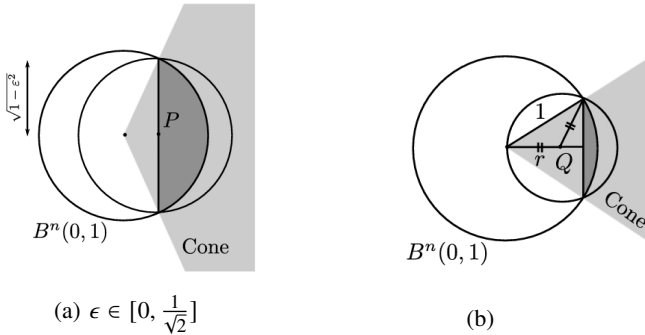
The following lemma follows from the volumn argument of spherical cap using ball (see figure 1a) [1]. Given a fixed unit vector $u \in \mathbb{S}^{d-1}$ and a parameter $\epsilon \in [0, 1]$, the spherical cap centered at u with parameter ϵ is defined as:

$$C(u, \epsilon) = \{v \in \mathbb{S}^{d-1} \mid \langle u, v \rangle \geq \epsilon\}$$

where $\mathbb{S}^{d-1}$ denotes the unit sphere in $\mathbb{R}^d$.

Lemma 1. Let $u, v \in \mathbb{R}^d$ be unit vectors chosen independently and uniformly at random. Then for all $\epsilon > 0$,

$$\mathbb{P}(\langle u, v \rangle \geq \epsilon) < e^{-\epsilon^2 d/2} \quad (2)$$



Let $\mathbb{P}$ be the uniform probability measure on the unit sphere $\mathbb{S}^{d-1}$. Draw a ball with radius r such that the spherical cone (the shadow area) lies in the ball. For $\epsilon \in [0, \frac{1}{\sqrt{2}}]$, take the ball. $\mathbb{P}(\langle u, v \rangle \geq \epsilon) \leq (1 - \epsilon^2)^{d/2} \leq e^{-\epsilon^2 d/2}$. For $\epsilon \in (\frac{1}{\sqrt{2}}, 1]$, take the ball with center Q and radius $\frac{1}{2\epsilon}$. Then $\mathbb{P}(\langle u, v \rangle \geq \epsilon) \leq (\frac{1}{2\epsilon})^d \leq e^{-\epsilon^2 d/2}$.

We can use Lemma 1 to show that for any fixed $\epsilon \in [0, 1]$, there are at least exponentially many almost orthogonal unit vectors in $\mathbb{R}^d$, whose pairwise inner product is $\leq \epsilon$. Picking m unit vectors $(v_i)_{i=1}^m$ in $\mathbb{R}^d$, and taking union bound for all pairs of

them, and choosing m such that the probability less than or equal to 1, we get

$$\mathbb{P}(\exists i \neq j, |\langle v_i, v_j \rangle| \geq \epsilon) < 2 \binom{m}{2} e^{-\epsilon^2 d/2} < m^2 e^{-\epsilon^2 d/2} \leq 1. \quad (3)$$

Choose $m = e^{\epsilon^2 d/4}$ such that the with positive probability $|\langle v_i, v_j \rangle| \leq \epsilon$ occurs. This gives us a lower bound of the maximum number of unit vectors in $\mathbb{R}^d$ with pairwise inner products in $[-\epsilon, \epsilon]$:

Theorem 2 (Lower Bound). For any $\epsilon > 0$, the maximum number m of unit vectors in $\mathbb{R}^d$ with pairwise inner products in $[-\epsilon, \epsilon]$ satisfies

$$m \geq e^{\frac{\epsilon^2 d}{4}}.$$

III. THRESHOLD REGIMES: HOW ϵ VALUES DETERMINE MAXIMUM CAPACITY

By the probabilistic argument (Equation 3), we can only assert that the lower bound of M is exponential in d , for fixed ϵ . When $\epsilon = \frac{c}{\sqrt{d}}$ for some constant c , $m \geq e^{\frac{\epsilon^2 d}{4}} = e^{\frac{c^2}{4}}$ is a constant, which indicates the lower bound is not tight. The argument becomes vacuous for finding the true behavior of M . This scaling of ϵ with dimension is precisely where our analysis needs to be more refined. Before we deal with the case of $\epsilon \sim \frac{A}{\sqrt{n}}$, we will first consider the case of fixed ϵ . We distinguish several cases of **fixed** ϵ in terms of d :

- If $\epsilon = 1$, m can be arbitrarily large.
- If ϵ is in $[\frac{1}{2}, 1)$, M is exponential in d . The upper bound of m is in form of $c(1 - \epsilon)^{-d/2}$ for some absolute constant c , which can be derived from volumn argument on spherical cap. We will prove it later.
- If ϵ is in $[\frac{1}{2\sqrt{d}}, \frac{1}{2}]$, M is exponential in dimension d . The upper bound of m is in form of $e^{c\epsilon^2(\log \frac{1}{\epsilon})d}$ for some absolute constant c , following from the result of Noga Alon's Theorem 9.3 on the high rank of perturbations of the identity matrix [2].
- If ϵ is in $[0, \frac{1}{2\sqrt{d}}]$, M is polynomial in dimension d . The minimum of ϵ that allows $m = d + 1$ is $\frac{1}{d}$ (in this case, those vectors form a regular simplex centered at the origin, which is also a equiangular tight frame).

The form $e^{c\epsilon^2(\log \frac{1}{\epsilon})d}$ as the **upper bound** of m , is actually works for ϵ in $[0, \frac{1}{2}]$, but it is tight only for $\epsilon \in [\frac{1}{2\sqrt{d}}, \frac{1}{2}]$ which can be interpreted as the following: For $\epsilon \in [0, \frac{1}{2\sqrt{d}}]$, taking $\epsilon = \frac{1}{4\sqrt{d}}$ as an example, $e^{c\epsilon^2(\log \frac{1}{\epsilon})d} = e^{\frac{c}{16d}(\log 4\sqrt{d})d} = e^{\frac{c}{32} \log 16d} = (16d)^{c/32}$, which is a polynomial of d . This demonstrates that one cannot remove $\log \frac{1}{\epsilon}$ from the exponent in the upper bound for ϵ in $[0, \frac{1}{2}]$, otherwise the upper bound becomes a constant i.e. e^c . The constant upper bound contradicts with an existing construction of vectors by number-theoretic method, which gives For $\epsilon \in [\frac{1}{2\sqrt{d}}, \frac{1}{2}]$, taking $\epsilon = \frac{1}{\sqrt{n}}$, $e^{c\epsilon^2(\log \frac{1}{\epsilon})d} = e^{c\frac{1}{n}(\log \sqrt{n})n} = e^{c\frac{1}{2} \log n} = (\sqrt{n})^{c/2}$, which is a polynomial of d .

For the proof of those cases, the bound of first and the third case is relatively easy to derive, while the second case

needs more effort. There are two proofs for the second one provided by Tao [3] and Alon [2]. Tao use volume packing argument for ϵ relatively large (in $[\frac{1}{2}, 1]$) [3]: For $\|v_i - v_j\|^2 = (1 - \epsilon^2) + 1 - \epsilon^2 = 2(1 - \epsilon)$. We set $r = \frac{1}{2}\sqrt{2(1 - \epsilon)}$. Then the open balls with radius r centered at each v_i are disjoint and they all belong to a larger ball with bounded radius $O(1)$. Hence $m \leq \frac{\text{Vol}(B(0, O(1)))}{\text{Vol}(B(v_i, r))} \leq (\frac{C}{r})^n = c^n$, for some absolute constant c .

A. For ϵ scaling with d

Now we consider $\epsilon = \frac{A}{\sqrt{n}}$ with $\frac{1}{2} \leq A \leq \frac{1}{2}\sqrt{n}$. The upper bound $e^{c\epsilon^2(\log \frac{1}{\epsilon})d} = (\frac{Cn}{A^2})^{CA^2}$ is tight by known constructions of error-correcting codes, see Tao's blog [3].

Theorem 3 (Cheap Kabatjanskii-Levenstein Bound). *Let $\{v_1, \dots, v_m\}$ be unit vectors in $\mathbb{R}^n$ such that $|\langle v_i, v_j \rangle| \leq An^{-1/2}$ for some $1/2 \leq A \leq \frac{1}{2}\sqrt{n}$. Then we have $m \leq (\frac{Cn}{A^2})^{CA^2}$ for some absolute constant C .*

For $0 \leq A < \frac{1}{2}$, the maximum number m of unit vectors in $\mathbb{R}^n$ with pairwise inner products bounded by $\frac{A}{\sqrt{n}}$ in absolute value satisfies $m = O(n)$. This result is essentially optimal, as demonstrated by the example of an orthonormal basis.

Lemma 2. *Let $\{v_1, \dots, v_m\}$ be unit vectors in $\mathbb{R}^n$ such that $|\langle v_i, v_j \rangle| \leq \frac{1}{2\sqrt{n}}$ for all distinct i, j . Then $m < 2n$.*

Proof. Suppose for contradiction that $m \geq 2n$. Consider the $2n \times 2n$ Gram matrix $G = (\langle v_i, v_j \rangle)_{1 \leq i, j \leq 2n}$. This matrix is real symmetric with rank at most n . If we subtract the identity matrix from G , the resulting matrix $G - I$ has an eigenvalue of -1 with multiplicity at least n .

Taking Hilbert-Schmidt norms, we conclude that

$$\sum_{1 \leq i, j \leq 2n: i \neq j} |\langle v_i, v_j \rangle|^2 \geq n. \quad (4)$$

However, by our hypothesis, the left-hand side is bounded above by

$$2n(2n - 1) \cdot \frac{1}{4n} = n - \frac{1}{2}, \quad (5)$$

which is less than n . This contradiction proves the lemma. $\square$

More generally, for $\epsilon = \frac{A}{\sqrt{n}}$ with $0 \leq A < \frac{1}{2}$, the maximum number of unit vectors with pairwise inner products at most ϵ in absolute value is $O(n)$.

For $\frac{1}{2}\sqrt{n} \leq A \leq \sqrt{n}$, using the same volume packing argument gives an upper bound $m \leq (1 - An^{-1/2})^{-d/2}$.

IV. EQUIANGULAR LINES

This section extend the problem to the case of $|\langle v_i, v_j \rangle| = \epsilon$, and $|\langle v_i, v_j \rangle| \approx \epsilon$.

A. Number of Approximately Equiangular Lines

Before proceeding, we introduce some definitions and notations. Sets $\{v_j\}$ of n unit vectors in $\mathbb{R}^d$ with a small maximum cross-correlation $M := \max_{j \neq k} |\langle v_j, v_k \rangle| \in [0, 1]$ have applications in wireless communication and coding theory. The lower bound for M is attained as follows.

Definition 1. A set $\{v_j\}$ of equal-norm vectors in $\mathbb{R}^d$ is *equiangular* if $\exists C \geq 0$ such that $|\langle v_j, v_k \rangle| = C$, $\forall j \neq k$. For a set of equiangular vectors of unit length, the constant is often written $C = \alpha = \cos \theta$, $\alpha \in [0, 1]$, $\theta \in [0, \pi/2]$, with both $C = \alpha$ and θ referred to as the angle between the vectors (or lines).

Definition 2 (Equiangular Lines). A set of n lines through the origin in $\mathbb{R}^d$ is said to be *equiangular* if the angle between any two distinct lines is the same.

Given a set of equiangular lines with common angle $\arccos \alpha$. By choosing a vector from each line, we can get a set of unit vectors with the property that the inner product of any two vectors equals to either α or $-\alpha$. This set of vectors is a special type of spherical code.

Definition 3 (Spherical L-Code). Let L be a subset of the interval $[-1, 1]$. A finite non-empty set C of unit vectors in Euclidean space $\mathbb{R}^n$ is called a *spherical L-code*, or for short an *L-code*, if $\langle x, y \rangle \in L$ for any pair of distinct vectors x, y in C .

Note that the set of vectors defined above is a $\{\alpha, -\alpha\}$ -code in $\mathbb{R}^d$. Conversely, given a $\{\alpha, -\alpha\}$ -code in $\mathbb{R}^d$, we can get a set of vectors, and the equiangular lines induced by those vectors do not change when we negate the vectors. Our problem is to find the maximum size of a $[-\epsilon, \epsilon]$ -code in $\mathbb{R}^d$, thus it is equivalent to the problem of finding the maximum number of approximately equiangular lines in $\mathbb{R}^d$. Theorem 5.2.1 from [4], providing a **lower bound** for the maximum number of approximately equiangular lines in $\mathbb{R}^d$ with angle $[\arccos(\alpha - \beta), \arccos(\alpha + \beta)]$.

Theorem 4 (Hoeffding's inequality). Let $X_1, \dots, X_n$ be independent centered random variables with $X_i \in [-1, 1]$ for all i . Let $X = \sum_{i=1}^n X_i$. Then for all $t > 0$,

$$\mathbb{P}(X \geq t\sqrt{n}) \leq e^{-t^2/2}. \quad (6)$$

Proof of theorem 4. We apply the Chernoff method, which uses the moment generating function to bound tail probabilities. For any $\lambda > 0$, by Markov's inequality:

$$\mathbb{P}(X \geq t\sqrt{n}) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t\sqrt{n}}) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t\sqrt{n}}}$$

For bounded random variables $X_i \in [-1, 1]$, we can show that $\mathbb{E}[e^{\lambda X_i}] \leq e^{\lambda^2/2}$. This follows from the convexity of the exponential function and Taylor expansion.

By independence, $\mathbb{E}[e^{\lambda X}] = \prod_{i=1}^n \mathbb{E}[e^{\lambda X_i}] \leq \prod_{i=1}^n e^{\lambda^2/2} = e^{n\lambda^2/2}$

$$\text{Therefore: } \mathbb{P}(X \geq t\sqrt{n}) \leq \frac{e^{n\lambda^2/2}}{e^{\lambda t\sqrt{n}}} = e^{n\lambda^2/2 - \lambda t\sqrt{n}}$$

Setting $\lambda = \frac{t}{\sqrt{n}}$ to minimize the bound: $\mathbb{P}(X \geq t\sqrt{n}) \leq e^{n \cdot t^2/2n - t^2/\sqrt{n} \cdot \sqrt{n}} = e^{t^2/2 - t^2} = e^{-t^2/2}$ $\square$

Theorem 5 (Exponentially many approximately equiangular vectors (Theorem 5.2.1) [4]). For every $\alpha \in (0, 1)$ and $\beta > 0$, there exists $c > 0$ such that for every n , one can find at least 2^{cn} unit vectors in $\mathbb{R}^d$ whose pairwise inner products all lie in $[\alpha - \beta, \alpha + \beta]$.

Proof. Let $\alpha \in (0, 1)$ and $\beta > 0$. Set $p = \frac{1+\sqrt{\alpha}}{2}$. For $i = 1, \dots, m$, let each $v_i \in \{-1, +1\}^d$ be chosen independently so that each coordinate is $+1$ with probability p and -1 with probability $1 - p$.

Then for $i \neq j$, the dot product $v_i \cdot v_j$ is a sum of d independent ± 1 -valued random variables each with mean α . Center each variable $(v_i^{(k)} \cdot v_j^{(k)} - \alpha)/(1 + \alpha)$, and choose $t = \frac{\beta\sqrt{d}}{1+\alpha}$. By Theorem 4, we have

$$\mathbb{P}(|v_i \cdot v_j - \alpha d| \geq \beta d) \leq 2e^{-\frac{\beta^2}{1+\alpha}d}.$$

A union bound over all pairs:

$$\mathbb{P}(\exists i \neq j, |v_i \cdot v_j - \alpha d| \geq \beta d) < m^2 e^{-\frac{\beta^2}{1+\alpha}d} < 1 \quad (7)$$

This equation shows that with positive probability, no pair $(i \neq j)$ has $|v_i \cdot v_j - \alpha d| \geq \epsilon d$, for some m at least 2^{cd} . Therefore, scaling each v_i by $\frac{1}{\sqrt{d}}$ yields $m = 2^{cd}$ unit vectors in $\mathbb{R}^d$ whose pairwise inner products lie in $[\alpha - \epsilon, \alpha + \epsilon]$. This completes the proof. $\square$

The proof of this lower bound is based on the probabilistic method, which is provided by Lemma 1. By setting $\alpha = \frac{\epsilon}{2}$ and $\beta = \frac{\epsilon}{2}$, we obtain the desired region of inner product $[0, \epsilon]$ for equiangular lines induced by a set of vectors, and consequently, the desired region of inner product $[-\epsilon, \epsilon]$ for the set of vectors.

This upper bound complements our lower bound and establishes that the maximum number of unit vectors in $\mathbb{R}^d$ with pairwise inner products in $[\alpha - \beta, \alpha + \beta]$ is $2^{\Theta(d)}$ for some constant $C_{\alpha, \beta}$ that depends only on α and β . However, this bound is independent of α and β . In previous section, we have similar bound depending on ϵ .

B. Number of Equiangular Lines

How about the we require cross-correlation is exactly ϵ ? How many unit vectors can we get. It is equivalent to find $N_\epsilon(d)$: the maximum number of lines in $\mathbb{R}^d$ with pairwise angles $\arccos \epsilon$. The answer is $N_\epsilon(d)$ is linear in d , and for large d it was proved to have a closed-form solution for some particular ϵ s by Zhao [5].

Without fixed cross-correlation, how many we can get? Let $N(d)$ be the maximum number of lines in $\mathbb{R}^d$ with pairwise equal angles. The answer is that $N(d)$ is quadratic in d .

$$cd^2 \leq N(d) \leq \binom{d+1}{2} \quad (8)$$

Welch bounds give a lower bound on how small the cross-correlation of a set of signals of unit energy can be.

Theorem 6 (Welch Bound). *Let $v_1, v_2, \dots, v_m$ be unit vectors in $\mathbb{R}^d$, $m > d$. For $\{v_1, \dots, v_m\}$ unit vectors in $\mathbb{R}^d$. Define the maximum cross-correlation of the set of vectors as $M := \max_{i \neq j} |\langle v_i, v_j \rangle|$. Then M^2 has a lower bound given by:*

$$M^2 \geq \frac{m-d}{d(m-1)} \quad (9)$$

Proof. Define G as the Gram matrix of the vectors $\{v_i\}_{i=1}^m$:

$$G = [v_i^T v_j]_{1 \leq i, j \leq m}.$$

Each v_i has unit length, so the diagonal entries of G are all 1, and thus $\text{Tr}(G) = m$. Since G is positive semidefinite with rank at most d , it has at most d positive eigenvalues $\lambda_1, \dots, \lambda_r$ (with $r \leq d$). By Cauchy-Schwarz,

$$(\text{Tr}(G))^2 = \left(\sum_{i=1}^r \lambda_i\right)^2 \leq r \sum_{i=1}^r \lambda_i^2 \leq d \sum_{i=1}^m \lambda_i^2.$$

The sum of squares of the Gram matrix elements equals its Frobenius norm squared:

$$\|G\|^2 = \sum_{i=1}^m \sum_{j=1}^m |v_i^T v_j|^2 = \sum_{i=1}^m \lambda_i^2.$$

Combining both,

$$\sum_{i=1}^m \sum_{j=1}^m |v_i^T v_j|^2 \geq \frac{(\text{Tr}(G))^2}{d} = \frac{m^2}{d}.$$

Separating the diagonal from the off-diagonal terms:

$$\sum_{i \neq j} |v_i^T v_j|^2 = \sum_{1 \leq i, j \leq m} |v_i^T v_j|^2 - \sum_{i=1}^m |v_i^T v_i|^2 \geq \frac{m^2}{d} - m = \frac{m(m-d)}{d}.$$

Since there are $m(m-1)$ off-diagonal terms and each is at most M^2 by definition of M ,

$$m(m-1)M^2 \geq \sum_{i \neq j} |v_i^T v_j|^2 \geq \frac{m(m-d)}{d}.$$

Therefore,

$$M^2 \geq \frac{m-d}{d(m-1)}.$$

$\square$

The Welch bound is achieved if and only if the vectors are ETF. The existence of ETF for any dimension is not completely understood. The paper [6] gives some conditions and constructions for ETF.

V. DISCUSSION

This 'simple' problem is actually very hard to solve. It remains open whether $\log \frac{1}{\epsilon}$ can be removed for fixed ϵ , which is highly related to the problem of removing the logarithm in Johnson-Lindenstrauss lemma. I'm not tootally understand many detail in the regimes of ϵ scaling with d . By the way, I learnt many new intersting things from this paper, such as the probabilistic method, the existence of ETF, and the connection between equiangular lines and spherical codes. The most interesting thing is that tight frames can be viewed as a orthogonal projection of a set of orthonormal basis(see figure 2): Every finite frame $\Phi = (f_j)_{j \in J}$ is a normalized tight frame iff it is the projection of orthonormal basis for $l(J)$.

Future work includes the following:

- 1) extend the result to the case of independent random variables with *arbitrary* distributions.

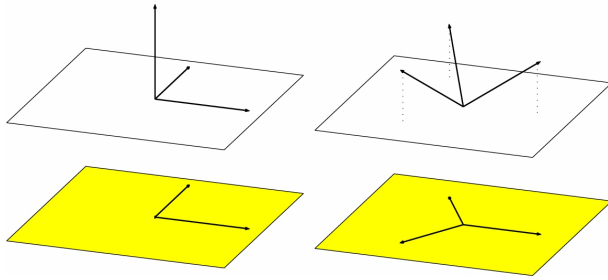


Fig. 2.4: The normalised tight frames $\{0, e_1, e_2\}$ and three equally spaced vectors obtained as the orthogonal projection of an orthonormal basis for $\mathbb{R}^3$ onto $\mathbb{R}^2$.

Fig. 2: Tight frames as orthogonal projections of orthonormal bases

- 2) explore the pseudo-randomness of the random variables, and how to generate approximately independent random variables from them.
- 3) discuss the case of *dependent* random variables and how to generate approximately independent random variables from them.
- 4) understand the number-theoretic method [3] for $\epsilon \ll \frac{1}{\sqrt{n}}$.
- 5) the behaviour of the absolute constant c in the upper bound of m ?
- 6) relation between **random** projection and tight frames.

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